

Laboratory assignment №7

Topic: : Building real-time web applications with SignalR

Goal: to gain hands-on experience integrating SignalR into a web application to enable real-time data exchange using SignalR.

Workflow:

Task.

Add SignalR to your MVC project developed in the previous labs. Implement real-time functionality according to the topic of your project (Table 1).

Number on group list	Topic	Feature
3	Job Portal	Instant notifications to candidates when a new job in their preferred category is posted.

Fig 1 – My topic

What functionality did I add and the general algorithm of work:

The candidate selects categories

when a new vacancy is created then:

- 1) A vacancy record is created
- 2) we get a list of user IDs who are subscribed to this category (a special service for such interactions)
- 3) Notifications are created in the database
- 4) A message is sent to these users in real time via SignalR.

On the client:

_Layout has JS that connects to the SignalR hub.

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Змн.	Арк.	№ докум.	Підпис	Дата							
Розроб.		Ліщинський В. В.			Звіт з лабораторної роботи				Лім.	Арк.	Аркушів
Перевір.		Українець М. О.								1	3
Керівник									ФІКТ, зр. ІПЗк-24-1		
Н. контр.											
Зав. каф.											

```

@RenderSection("Scripts", required: false)
<script src="~/lib/bootstrap/js/bootstrap.bundle.min.js"></script>

<script>
    const connection = new signalR.HubConnectionBuilder()
        .withUrl("/notificationHub")
        .build();

    connection.on("ReceiveNotification", (message) => {
        console.log("Нова вакансія: " + message);

        const notifDiv = document.getElementById("notificationArea");
        if (notifDiv) {
            notifDiv.innerHTML = `<div class="alert alert-info alert-dismissible fade show" role="a

```

Fig 2 – On the client

```

namespace JobPortal.Components
{
    1 reference
    public class UserIdProvider : IUserIdProvider
    {
        0 references
        public string GetUserId(HubConnectionContext connection)
        {
            return connection.User?.FindFirst(ClaimTypes.NameIdentifier)?.Value;
        }
    }
}

```

Fig 3 - UserIdProvider

```

[2025-05-24T18:38:13.103Z]      signalr.min.js:1
Information: Normalizing '/notificationHub' to '
http://localhost:5000/notificationHub'.

▶ Fetch finished loading: POST "      signalr.min.js:1
http://localhost:5000/notificationHub/negotiate?ne...
".

▶ XHR finished loading: GET "      browserLink:21
http://localhost:50887/2a95a2d.../browserLinkSignalR...
".

[2025-05-24T18:38:13.272Z]      signalr.min.js:1
Information: WebSocket connected to ws:
//localhost:5000/notificationHub?
id=GqP7wLZkae10rg9Tfge-2A.
>

```

Fig 4 – SignalR connect

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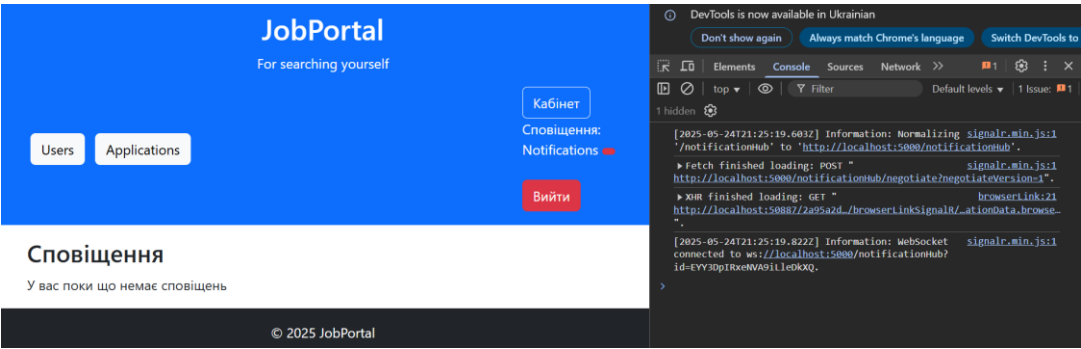


Fig 5 – Notification counter in frontend

Description

I am admin

Категорії, які вас цікавлять:

☐ ІТ, комп'ютери

☐ Бухгалтерія, аудит

☐ Готельно-ресторанний бізнес, туризм

☐ Дизайн, творчість

☐ ЗМІ, видавництво

☐ Краса, фітнес, спорт

☒ Нерухомість

Зберегти зміни

Fig 6 – Selected categories in profile

Conclusions: During the laboratory work, I gained hands-on experience integrating SignalR into a web application to enable real-time data exchange using SignalR.